Higher National Diploma in Engineering (C \ E \ M BSE)

First Year, First Semester Examination -- 2017

MA 1101: Engineering Mathematics

Question 1 (Q1.)

a) The matrix A is defined as:

$$A = \begin{pmatrix} -2 & 5 & 4 \\ 5 & 7 & 5 \\ 4 & 5 & -2 \end{pmatrix}$$

Find:

i) the **characteristic equation** of the matrix A. (5 Marks)

ii) **Eigen values** and the corresponding **Eigen vectors** of the matrix A. (12 Marks)

b) Write down the **quadratic form** corresponding to the matrix A:

$$A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

(8 Marks)

Question 2 (Q2.)

a) Test the **convergence** of the following series:

$$\frac{1}{123} + \frac{3}{234} + \frac{5}{345} + _\ldots\ldots\ldots\ldots\ldots\ldots$$

(10 Marks)

b) Test for **convergence** of the series:

$$\frac{x}{2} + x^2 + \frac{3x^3}{8} + x^4 + \frac{25x^5}{32} + _\ldots\ldots\ldots\ldots\ldots\ldots$$

(15 Marks)

Question 2 (Q2.)

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(15 Marks)

Question 3 (Q3.)

a) If $ax^2+2hxy+by^2=1$, **Prove that** $\frac{d^3y}{dx^2}=\frac{h^2-ab}{(hx+by)^3}$. (10 Marks)

b) If $u=\sec^{-1}\left(\frac{x^3-y^3}{x+y}\right)$, Using **Euler's theorem on homogeneous function**, **Show that** $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=2\ \cot u$. (7 Marks)

c) Find $\frac{\partial u}{\partial r}$ and $\frac{\partial u}{\partial \theta}$ If $U=e^{r\cos\theta}\cos(r\sin\theta)$. (8 Marks)

Question 4 (Q4.)

a) If $x=r\ \sin\ \theta\ \cos\ \phi$, $y=r\ \sin\ \theta\ \sin\ \phi$, $z=r\ \cos\ \theta$, **Show that** $\frac{\partial(r,\theta,\phi)}{\partial(x,y,z)}=\frac{1}{r^3\sin\theta}$. (12 Marks)

b) **Evaluate** $\lim_{x\to 2}\frac{x^x-x}{x-1-\log x}$. (8 Marks)

c) **Evaluate** $\lim_{(x,y)\to(1,2)}\frac{3x^3y}{x^3+y^3+5}$. (5 Marks)

Question 5 (Q5.)

a) **Evaluate** $\iint_D xy\,dx\,dy$ where D is the region bounded by the curve $x=2-y$, $x=y^2$, $y=0$ and $y=1$. (10 Marks)

b) **Evaluate** $\int_0^{2\pi}\int_0^y\int_0^6(xy+yz+zx)\,dz\,dy\,dx$. (5 Marks)

c) Find the **volume bounded by the cylinder** $x^2+y^2=4$ and the planes $y+z=3$ and $z=0$. (10 Marks)